

concurrency model → predicted minimal-edit win

single event loop sync offload stalls everything	event loop + pool stalls one loop of N	thread / goroutine pool pool overlaps the rest	proxy + agent agent offloads async	process-per-connection OS overlaps connections free
2.5-3.0×	2.7-2.9×	3.0-3.5× (0 async code)	2.1× (C agent)	intra-query pipelining

Redis, Node

nginx, memcached

Apache, Go

HAProxy

Postgres, MariaDB

Win grows with offload weight; it pays only when the offload outweighs per-request CPU.